MILITARY SPECIFICATION

TEST SET, INDICATOR

AN/APM-433

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - The test equipment covered by this specification shall be for shop use and shall provide an accurate means of testing the ID-2206/APN-194(V) Height Indicator when used with the AN/APM-403 Test Set, Altimeter Set.

1.2 Classification - The test equipment covered by this specification shall consist of the following item:

Item	Type Designation	Applicable Paragraph
Test Set	AN/APM-433	3.5.1

The test set shall be Type I, Class 4, and shall be housed in an enclosure that is Style C, Color Y, as defined in MIL-T-28800.

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in paragraph 6.10.

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents - The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Military

MIL-C-5541	Coatings, Chem. Conversion
MIL-P-15024	Plates, Tags, and Bands for Identification of Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-D-18300	Design Data for Avionics Equipment
MIL-N-18307	Nomenclature and Nameplate for Aeronautical Electronic and Associated Equipment
MIL-T-28800	Test Equipment for Use with Electronic and Electrical Equipment, General Specification for

STANDARDS

Military

MIL-STD-108	Definition and Requirement/Enclosure for Electrical Equipment
MIL-STD-415	Test Points and Test Facilities for Electronic Systems and Associated Equipment
MIL-STD-461	EMI Characteristics, Requirement for Equipment
MIL-STD-462	EMI Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First Article - When specified, a sample shall be subjected to first article inspection.

3.2 Parts and Material - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(a) Parts and materials requirements shall conform to MIL-T-28800.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.

(b) Nonrepairable subassemblies, as outlined in MIL-T-28800, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4)

(c) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-T-28800.

3.2.2 Electron Devices - Electron tubes, transistors, and diodes shall be chosen and applied as outlined in MIL-T-28800.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of MIL-T-28800 for design, construction and workmanship except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, including cables, shall be a minimum consistent with good design and shall not exceed 30 pounds.

3.3.2 Reliability

3.3.2.1 Operational Stability - The equipment shall operate in a shop environment with satisfactory performance, continuously or intermittently, for a period of at least 10,000 hours, without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a minimum total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment, and the wear-out life of such parts, shall be determined by the contractor and submitted to the procuring agency.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment shall have a minimum of 69,000 hours of mean (operating) time between failures.

3.3.3 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with MIL-T-28800. Insofar as is practicable, all connectors on the front panel shall be mounted along the lower edge of the panel and in no case shall their location be such that the connecting cables will interfere with the operating controls. Identification markings for external cables shall be on metal tags attached to the cables. Individual wire identification shall not be required.

Source: https://assist.dla.mil -- Downloaded: 202640-23T11:12Z
Check the source to verify that this is the currentversion before use.

3.3.4 Interchangeability - The equipment shall meet the interchangeability requirements as defined in MIL-T-28800. All removable elements shall be so configured as to preclude possibility of improper assembly.

3.3.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of MIL-STD-461 and MIL-STD-462.

3.3.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-T-28800. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance in accordance with MIL-STD-415.

3.3.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307 and MIL-P-15024.

3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Temperature	Room ambient (30° ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient
Input Power Voltage	27.0 ± 1.0 V DC

3.3.9 Service Conditions - The test equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these as specified in MIL-T-28800 for Type I Class 4 equipment.

3.3.10 Primary Input Power Requirements - The test equipment shall be capable of satisfactory operation when supplied input power having characteristics as defined in MIL-T-28800. The power required shall be provided by the AN/APM-403 Test Set and shall not exceed the specified amounts:

DC Power, 27.0 ± 2 V, 6 watts

3.3.11 Voltage Protection - The equipment shall be capable of withstanding input power variations as imposed by abnormal electric system operations as defined in MIL-STD-704.

3.3.12 Marking - The test equipment shall be marked and identified in accordance with MIL-T-28800 except that individual wire identification is not required and external cables shall be indentified by metal tags attached to them.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.

3.4 Performance - The performance of the equipment shall be as set forth below. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variation will be specified.

3.5 Detail Requirements

3.5.1 Test Set, Indicator, AN/APM-433 - The test set shall meet the following requirements:

3.5.1.1 Function - The test set shall provide the stimulus circuits, measurement test points and controls required for testing of the Altimeter Height Indicator indicated in Table I when used in conjunction with the AN/APM-403 Altimeter Set Test Set.

3.5.1.2 Form Factor - The test set shall be housed in a watertight combination case of overall dimensions (viewed from in-use position) not to exceed 16 inches wide by 11 inches high by 12 inches deep.

3.5.1.3 Weight - The weight of the test set shall not exceed 30 pounds, including the lid and accessory indicator cable.

3.5.1.4 Content - The test set shall contain the following assemblies:

(a) Control Panel
(b) Cable

3.5.1.4.1 Control Panel Assembly - The Control Panel Assembly shall provide the necessary controls and circuits to conduct performance tests on the Height Indicator listed in Table I when used in conjunction with the AN/APM-403 Altimeter Set Test Set. Supplemental instrumentation is required as indicated in paragraph 6.10. The Control Panel Assembly shall have maximum dimensions of 14.9 inches wide by 8.9 inches high by 1.25 inches deep behind the mounting surface. Control and circuit requirements are as follows:

3.5.1.4.1.1 Power Control - This shall consist of a switch to control power inputs to the unit under test (UUT). The switch shall allow selection of either UUT integral lighting with all other UUT power inputs interrupted or normal operation with all power inputs applied.

3.5.1.4.1.2 Visual Warning Indicators - Indicator lights shall be provided for the output of the UUT Visual Warning circuit. The indicators shall consist of three parallel-connected MS18235-3CN or equivalent lamps.

3.5.1.4.1.3 Reliability Signal Control - The test set shall provide a Reliability Signal logic input to the UUT with the logic states selectable from the front panel. The voltage levels for the logic states shall be as follows:

ON	4.5 ± 1.0 VDC
OFF	$0.0 \pm \begin{matrix} 0.0 \\ 0.5 \end{matrix}$ VDC

Source: https://assist.dla.mil -- Downloaded: 202640-23T11:12Z
Check the source to verify that this is the current version before use.

3.5.1.4.1.4 Weight on Wheels Signal Control - The test set shall provide logic input signals to the UUT Weight on Wheels logic circuit with the logic states selectable from the front panel. The voltage levels for the logic states shall be as follows:

ON	$0.0 \pm \begin{matrix} 0.5 \\ 0.0 \end{matrix}$ VDC
OFF	+4.0 to +29.0 VDC or open circuit

3.5.1.4.1.5 Landing Gear Down and Locked No. 1, 2, 3 Signal Controls - The test set shall provide separate logic input signals to each of the three UUT Landing Gear logic circuits with the logic states selectable from the front panel. The voltage levels for the logic states shall be the same as those specified in paragraph 3.5.1.4.1.4.

3.5.1.4.1.6 High, Low, and Variable Altitude Inhibit Signal Controls - The test set shall provide separate logic input signals to each of the three UUT Altitude Inhibit logic circuits with the logic states selectable from the front panel. The voltage levels of the logic states shall be the same of those specified in paragraph 3.5.1.4.1.4.

3.5.1.4.1.7 AFCS Coupler Engage Signal Control - The test set shall provide a logic signal to the UUT AFCS Coupler Engage logic circuit with the logic states selectable from the front panel. The voltage levels for the logic states shall be as follows:

ON	25.0 ± 4.0 VDC
OFF	$0.0 \pm \begin{matrix} 0.5 \\ 0.0 \end{matrix}$ VDC

3.5.1.4.1.8 Reliability Fail Signal Control - The test set shall provide a signal (the Rel Fail signal) to the UUT Altitude Range input that simulates range voltages for altimeter loss-of-track and in-track conditions. Selection of either the altitude range input provided by the AN/APM-403 Altimeter Set Test Set or the test input from this test set shall be made from the tester front panel. The Rel Fail signal shall be less than 19.500 VDC when the Reliability Signal (3.5.1.4.1.3) is ON and 24.000 $\pm$ 1.000 VDC when the Reliability Signal is OFF.

3.5.1.4.1.9 Aural Warning Load Select Control - The test set shall provide a load to the UUT Aural Warning Circuitry. A control on the front panel shall allow selection of either the 1.2 $\pm$ 5% kilohm load in the AN/APM-403 test set or the 600 $\pm$ 5% ohm load in this test set.

3.5.1.4.1.10 Test Points - The following test points shall be provided for monitoring UUT input and output signals:

Reference Designation	Signal Name	Corresponding Connector-Pin
TP1	Signal Ground	J1-Y
TP2	Servo Drive	J1-Z
TP3	Altitude Range Signal	J1-j

Source: https://assist.dla.mil -- Downloaded: 2026l0-23T11:12Z
Check the source to verify that this is the currentversion before use.

3.5.1.4.1.10 (Continued)

Reference Designation	Signal Name	Corresponding Connector-Pin
TP4	A/C 28 VDC	J1-T
TP5	Ground	J1-S
TP6	No. 1 Landing Gear Down and Locked	J1-R
TP7	No. 2 Landing Gear Down and Locked	J1-P
TP8	No. 3 Landing Gear Down and Locked	J1-N
TP9	Weight on Wheels	J1-M
TP10	High Altitude Inhibit	J1-L
TP11	Low Altitude Inhibit	J1-K
TP12	Variable Altitude Inhibit	J1-f
TP13	AFCS Coupler Engage	J1-e
TP14	Reliability Signal	J1-d
TP15	Visual Warning Output	J1-m
TP16	Aural Warning Output	J1-p
TP17	Linear Range	J1-g

3.5.1.4.2 Cable Assembly - The LAMPS Indicator Cable Assembly, reference designation W1, shall provide the interface between the test set and the UUT.

3.5.1.4.2.1. Connector Identification - The cable shall have three connectors, as follows:

Reference Designation	Plug Type	Function
P1	MS3116E22-55P	Connect to AN/APM-433 Test Set J1
P2	MS3116E14-18S	Connect to ID-2206 Indicator J1301
P3	MS3116E14-18SX	Connect to ID-2206 Indicator J1302.

3.5.1.4.2.2 Physical Characteristics - The cable shall be 48 + 4 inches long. It shall branch out to the two indicator connectors 36 + 2 inches from the P1 end of the cable. The cable shall be fabricated with a "heat shrink" type or vinyl outer jacket. The cable and each connector shall be identified with a metal tag.

3.5.1.5 Controls

Control	Operation	Monitor Test Point	Function
POWER	4PDT Toggle Switch	TP4 (27 VDC)	NORMAL - All UUT operational and lighting power applied.
			INTEG LGTNG ONLY - All UUT power inputs except for Integral Lighting inputs are interrupted

Source: https://assist.dla.mil -- Downloaded: 202610-23T11:12Z
Check the source to verify that this is the current version before use.

3.5.1.5 Controls (Continued)

Control	Operation	Monitor Test Point	Function
AFCS CPLR ENG	SP3T Toggle Switch	TP13	OFF - Applies ground to UUT AFCS Coupler Engage circuitry. OPEN - Applies open circuit to AFCS Coupler Engage circuitry. ON - Applies 27 VDC to AFCS Coupler Engage circuitry
WGT ON WHLS	SP3T Toggle Switch	TP9	OFF - Applies 27 VDC to UUT Weight on Wheels circuitry Open - Applies open circuit to UUT. ON - Applies ground to UUT.
HIGH ALT INHIB	SP3T Toggle Switch	TP10	OFF - Applies 27 VDC to UUT High Altitude Inhibit circuitry. OPEN - Applies open circuit to UUT. ON - Applies ground to UUT.
LOW ALT INHIB	SP3T Toggle Switch	TP11	OFF - Applies 27 VDC to UUT Low Altitude Inhibit circuitry. OPEN - Applies open circuit to UUT. ON - Applies ground to UUT.
VAR ALT INHIB	SP3T Toggle Switch	TP12	OFF - Applies 27 VDC to UUT Variable Altitude Inhibit circuitry OPEN - Applies open circuit to UUT. ON - Applies ground to UUT.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.

3.5.1.5 Controls (Continued)

Control	Operation	Monitor Test Point	Function
NO. 1 LNDNG GEAR D&L	SP3T Toggle Switch	TP6	OFF - Applies 27 VDC to UUT No. 1 Landing Gear Down and Locked circuitry.
			OPEN - Applies open circuit to UUT.
			ON - Applies ground to UUT.
NO. 2 LNDNG GEAR D&L	SP3T Toggle Switch	TP7	OFF - Applies 27 VDC to UUT No. 2 Landing Gear Down and Locked circuitry.
			OPEN - Applies open circuit to UUT.
			ON - Applies ground to UUT.
NO. 3 LNDNG GEAR D&L	SP3T Toggle Switch	TP8	OFF - Applies 27 VDC to UUT No. 3 Landing Gear Down and Locked circuitry.
			OPEN - Applies open circuit to UUT.
			ON - Applies ground to UUT.
REL SIG	SP3T Toggle Switch	TP14	OFF - Applies ground to UUT Reliability Signal input.
			OPEN - Applies open circuit to UUT.
			ON - Applies 4.5 VDC to UUT.
AURAL WARN LOAD	SPDT Toggle Switch	TP16	600 - Applies 600 ohm load to UUT Aural Warning Output circuitry.
			1.2K - Provides continuity from UUT Aural Warning Output to 1200 ohm load in the AN/APM-403 Altimeter Set Test Set.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.

3.5.1.5 Controls (Continued)

Control	Operation	Monitor Test Point	Function
REL FAIL	SPDT Toggle Switch	TP3	NORMAL - Provides for application of the AN/APM-403 Altitude Range Signal to the UUT. TEST -Applies signl corrre-sponding to reliable or unreliable operation as controlled by the REL SIG switch, to the UUT Altitude Range Signal input.

3.5.1.6 Electrical Connections - Connections to external circuits shall be provided by Control Panel Assembly connectors as follows:

Reference Designation	Plug Type	Function
J1301	MS3124E14-18P	AN/APM-403 Interface Connection
J1302	MS3124E14-18PX	AN/APM-403 Interface Connection
J1	MS3124E22-55S	Indicator Connection

3.5.2 Case - The test set shall be provided with a combination case which shall conform to MIL-T-28800 requirements for a style C case. The case shall be watertight and conform to MIL-STD-108.

3.5.2.1 Size - The case shall be 16 inches wide X 10.75 inches high X 11.35 inches deep.

3.5.2.2 Material - The case shall be constructed of aluminum that is strong enough to meet the requirements of MIL-T-28800 for a combination case.

3.5.2.3 Finish - The finish on the case shall be in accordance with MIL-T-28800B paragraph 3.3.4.47 Color Y except as follows:

(a) Panel mounting surface to be conductive chromate per MIL-C-5541, four sides.

(b) All inside surfaces below panel mounting surface to be chromate per MIL-C-5541.

3.5.2.4 Stowage - The case shall provide stowage space for cables, operating instructions and schematics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.

3.5.3 Cable Assemblies - A cable assembly shall be provided as follows:

Reference Designation	Plug Type	Function
W1	N/A	Interface AN/APM-433 Test Set to ID-2206 Indicator
W1-P1	MS3116E22-55P	Connect to AN/AMP-433 Test Set J1
W1-P2	MS3116E14-18S	Connect to ID-2206 Indicator J1301
W1-P3	MS3116E14-18SX	Connect to ID-2206 Indicator J1302

3.6 Design Data - Design data shall be in accordance with the requirements of MIL-D-18300. (See 6.7)

4.0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2.1 Classification of inspections - The inspection requirements specified herein are classified as follows:

(a) First article inspection (See 4.3)

(b) Quality conformance inspection (See 4.4)

4.3 First Article Inspection - First article inspection shall be made on an equipment representative of the production equipments to be supplied under the contract. First article inspection shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedures of 4.5. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.3.1 First article inspection Data - Data collected in conducting these tests shall be supplied to the procuring agency for review and approval.

Source: https://assist.dla.mil -- Downloaded: 202610-23T11:12Z
Check the source to verify that this is the current version before use.

4.3.2 Scope of Tests - First article inspection shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. First article inspection shall include the following environmental tests in accordance with MIL-T-28800: temperature, humidity, altitude, vibration, and salt-sea atmosphere followed by a bench test.

4.3.3 First article inspection approval - Approval of first article inspection units shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the first article inspection sample. Prefabrication of production equipment prior to the approval of the first article inspection sample is at the contractor's own risk.

4.3.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved first article inspection sample unless otherwise specified. Each equipment shall be capable of successfully passing the same tests as imposed on the first article inspection sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.4 Quality Conformance Inspection - The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance inspection. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the services of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all quality conformance inspection. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance inspection shall consist of the following:

(a) Individual Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(a) Examination of Product
(b) Operational Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

Source: https://assist.dla.mil -- Downloaded: 202640-23T11:12Z
Check the source to verify that this is the currentversion before use.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.5 Test Procedures - The procedures used for conducting first article inpection and quality conformance inspection shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.6 Reconditioning of Tested Equipment - Equipment which has been subjected to qualification tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.7 Presubmission Testing - No item, part of complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

6. NOTES

6.1 Intended Use - The equipment defined herein is for detailed testing of the ID-2206/APN-194(V) Height Indicator, Table I, when used in conjunction with the AN/APM-403 Altimeter Set Test Set.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltage levels supplied by the AN/APM-403 Test Set. These data are to be used in testing the equipment at installation points for compliance with the minimum acceptance standard of performance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity or design, performance, ease of maintenance, or reliability will result from the use of materials, parts, and processes other than those specified in MIL-T-28800, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by adequate supporting information.

6.4 Nonrepairable Subassemblies - As a general rule, nonrepairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts, tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. The nonrepairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 The Parenthesis (*) - The parenthesis (*) when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.7. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.6 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1).

(c) Data requirements (see 3.6).

6.7 Data Item Discriptions - When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID NO.
3.6	Report, First Article (Preproduction) Test	UDI-T-21349

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.8 First Article - When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the documents first article.

6.9 Asterisk - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.10 Associated Equipment - The test set shall operate with the following associated equipment or equivalent, which shall not be supplied with the test set:

Item	Mfgs Code - Model	FSN
Test Set, Altimeter Set	94580-AN/APM-403	4920-01-026-2360

6.11 Supplemental Equipment - The test set shall operate with the following supplemental equipment or equivalent which shall not be supplied with the test set:

Item	Mfgs Code - Model	FSN
Oscilloscope	80058-AN/USM-281C	6G6625-106-9622
Digital Voltmeter	89536-8300A-01-02	6G6625-470-2721-BP
Multimeter	80058-AN/USM-311	2Z6625-160-1301

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.

TABLE I

RADAR ALTIMETER COMPONENTS

	AN/APN-194(V)	
Receiver-Transmitter	RT-1015 RT-1042	
Height Indicator	ID-2206 ID-1760A ID-1768 ID-1768A ID-1811	(Note 1)
Interference Blanker	MX-9132A	
Antenna	AS-2595 AS-2728 AS-2741 AS-2742	

Note 1: Only this item tested with the AN/APM-433 Test Set.

Preparing Activity
Navy - AS
(Project No. 6625-N812)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

OFFICIAL BUSINESS
ENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards
Department
ESSD (Code 93)
Lakehurst, NJ 08733

OLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-T-85601 (AS), Test Set, Indicator AN/APM-433

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76

Replaces edition of 1 Jan 72 which may be used.

S/N 0102-LF-001-4260

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:12Z
Check the source to verify that this is the current version before use.